



BAULKHAM HILLS HIGH SCHOOL

Half -Yearly 2013
YEAR 11 TASK 1

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-14
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Exam consists of 5 pages.

This paper consists of TWO sections.

Section 1 – Page 2 (10 marks)

Questions 1-10

- Attempt Question 1-10
- Answer on answer sheet provided

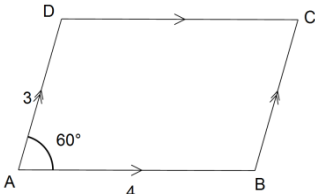
Section II – Pages 3-5 (60 marks)

- Attempt questions 11-14

Section I - 10 marks**Use the multiple choice answer sheet for question 1-10**

1. $x - 4$ is a factor of
 (A) $x^2 - 4$ (B) $x^2 + 4$ (C) $4x^2 - 16$ (D) $x^2 - 16$

2. The solution of $6 - \frac{x}{3} > x$ is
 (A) $9 > x$ (B) $x < \frac{3}{2}$ (C) $x < 4.5$ (D) $x > \frac{18}{4}$

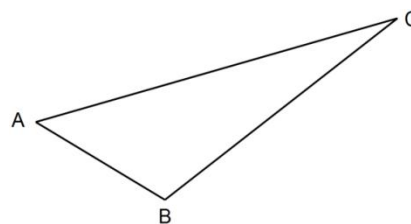
3.  The area of the parallelogram $ABCD$ is
 (A) $6 \times \frac{\sqrt{3}}{2}$ (B) $6\sqrt{3}$
 (C) 6 (D) 12

4. Which statement is correct?
 (A) $\tan 330^\circ = -\cot 30^\circ$ (B) $\tan 330^\circ = \tan 30^\circ$
 (C) $\tan 330^\circ = -\tan 30^\circ$ (D) $\tan 330^\circ = \cot 30^\circ$

5. The range of $f(x) = \sqrt{9 - x^2}$ is
 (A) all real y (B) $-3 \leq y \leq 3$ (C) $0 \leq y \leq 3$ (D) $0 \geq y \geq -3$

6. The bearing of B from A is 150° , C from B is 070° .
 What is the bearing of B from C

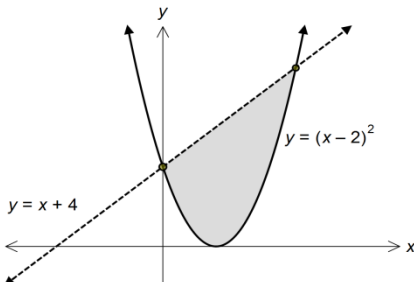
- (A) 030° (B) 250°
 (C) 110° (D) 280°



7. $f(x) = (x + 4)^2$ is
 (A) an even function (B) Neither an even nor odd function
 (C) an odd function (D) it's not a function

8. If $g(x) = \frac{x}{x^2+1}$ what is $g\left(\frac{1}{x}\right)$ equal to?
 (A) $\frac{x^2+1}{x}$ (B) $\frac{x^2}{x^2+1}$ (C) $\frac{x}{1+x^2}$ (D) $\frac{x^2}{x+x^2}$

9. $\left(x - \frac{1}{x}\right)^2 =$
 (A) $x^2 - \frac{1}{x^2}$ (B) $x^2 - 2 + \frac{1}{x^2}$ (C) $x^2 + 2 + \frac{1}{x^2}$ (D) $x^2 + \frac{1}{x^2}$

10.  The shaded region is represented simultaneously by the following inequalities
 (A) $y \leq (x - 2)^2$ and $y < x + 4$
 (B) $y \geq (x - 2)^2$ and $y > x + 4$
 (C) $y \geq (x - 2)^2$ and $y < x + 4$
 (D) $y \leq (x - 2)^2$ and $y > x + 4$

End of Section 1

Section II – Extended Response

Attempt questions 11-14. All necessary working should be shown in every question.

Question 11 (15 marks)		Marks
a)	Simplify $(4x - 1)^2 - 3(x - 5)$	2
b)	Evaluate correct to 3 significant figures $\frac{14.951 - 11.3}{\sqrt[3]{72}}$	2
c)	Solve $\begin{aligned} a + 3b &= 1 \\ a + 53b &= 101 \end{aligned}$	2
d)	Rationalise the denominator $\frac{2 + \sqrt{3}}{3 - 2\sqrt{3}}$	2
e)	Factorise and simplify $\frac{x^2 + x}{x^2 - 9} \times \frac{x^3 - 27}{x^2 - x - 2}$	3
f)	Consider the function $f(x) = \begin{cases} -3 & \text{if } x \leq -2 \\ x^2 & \text{if } -2 < x < 2 \\ 2x - 3 & \text{if } x \geq 2 \end{cases}$ (i) Evaluate $f(-1) + f(1)$ (ii) Find $f(a^2 + 2)$ (iii) Sketch the graph of $f(x)$	1 1 2
Question 12 (15 marks)		
a)	If $G(x) = 4x - x^2$ (i) Find the value of $G(-3)$ (ii) Find the values of x for which $G(x) = 0$ (iii) Sketch the graph of $G(x)$ showing all intercepts and the vertex.	1 1 2
b)	Find all values of θ such that $2 \cos^2 \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$	2
c)	(i) Sketch $f(x) = \frac{3}{x-1} - 1$ showing all intercepts	3
	(ii) Find the domain and range of $f(x)$	2
d)	If $\sin \theta = -\frac{1}{5}$ and $\tan \theta > 0$, find the exact value of $\cos \theta$	2
e)	Given that $(2\sqrt{3} + 1)(3 - \sqrt{3}) = a\sqrt{3} + b$, find a and b	2

Question 13 (15 marks)	Marks
a) Not to scale Find the exact value of x	1
b) Find the length AB to 2 decimal places.	2
c) Not to scale In the diagram below the lines $3x + y - 9 = 0$ and $4x - y + 16 = 0$ intersect at B. The point A is $(3,0)$ and C is $(-2,8)$ (i) Show that the line AC has the equation $8x + 5y - 24 = 0$ (ii) Show that the point B has coordinates $(-1,12)$ (iii) Find the equation of a line passing through B and perpendicular to AC. Answer in general form.	2 1 2
d) State whether the function $y = \frac{1}{x^3}$ is even, odd or neither. Justify your answer.	2
e) Simplify $\frac{2}{3-\sqrt{2}} + \frac{5}{3+\sqrt{2}}$	2
f) Write $\frac{a^{2x}}{a^{2y}} \times \frac{a^y}{a^2}$ in the form a^k	1
g) Simplify $\frac{y^{-1}+y}{1+y^2}$	2

Question 14 (15 marks)	Marks
a) Prove the identity $(\cos \alpha + \cot \alpha) \sec \alpha = 1 + \operatorname{cosec} \alpha$	2
b) Given $y = x^2$ and $y = 4x - 3 $ (i) Find the points of intersection of the two graphs. 3 (ii) On the same number plane, sketch $y = 4x - 3$ and $y = x^2$ showing the points of intersection. 2 (iii) Hence or otherwise solve: 1 $ 4x - 3 \geq x^2$	
c) The diagram shows an isosceles $\triangle ABC$ with $AC = BC$, $AB = 2\text{cm}$, $\angle BAC = 75^\circ$ and $\angle ACB = 30^\circ$. (i) Show that the exact value of $AC = \frac{2}{\sqrt{2-\sqrt{3}}}$ 2 (ii) Hence show that the exact value of $\sin 75^\circ = \frac{1}{2\sqrt{2-\sqrt{3}}}$ 2 (iii) Find the exact area of triangle ABC 1 (iv) Using part (ii) or otherwise show that the exact value of $\cos 75^\circ = \frac{\sqrt{2-\sqrt{3}}}{2}$ 2	

End of Exam



BAULKHAM HILLS HIGH SCHOOL

YEAR 11 HALF-YEARLY ASSESSMENT

2013

Name: Answer

Teacher: _____

Mathematics

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

- Start Here →
- | | |
|---|--|
| 1. A ☐ B ☐ C ☐ D ☒ | 6. A ☐ B ☒ C ☐ D ☐ |
| 2. A ☐ B ☐ C ☒ D ☐ | 7. A ☐ B ☒ C ☐ D ☐ |
| 3. A ☐ B ☒ C ☐ D ☐ | 8. A ☐ B ☐ C ☒ D ☐ |
| 4. A ☐ B ☐ C ☒ D ☐ | 9. A ☐ B ☒ C ☐ D ☐ |
| 5. A ☐ B ☐ C ☒ D ☐ | 10. A ☐ B ☐ C ☒ D ☐ |

~Q14 - page 1~

Question 11

a) $(4x-1)^2 - 3(x-5) = 16x^2 - 8x + 1 - 3x + 15$
 $= 16x^2 - 11x + 16$ ✓

b) $0.8776... = 0.878$ ✓

c) $a + 3b = 1$ $\therefore a + 6 = 1$

(2) $a + 53b = 101$

(2) - (1) $50b = 100 \therefore b = 2 \therefore a = -5$

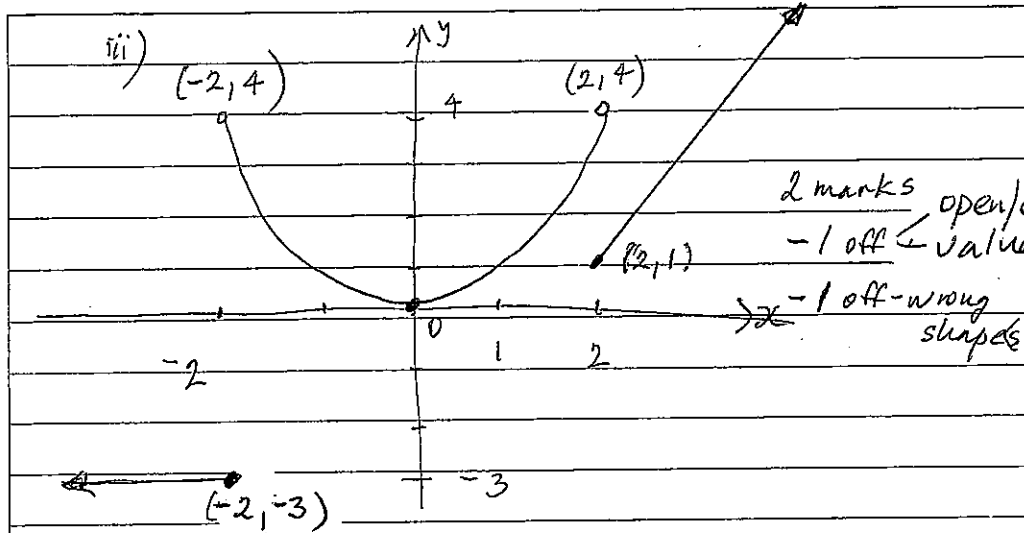
d) $\frac{2+\sqrt{3}}{3-2\sqrt{3}} \times \frac{3+2\sqrt{3}}{3+2\sqrt{3}} = \frac{6+3\sqrt{3}+4\sqrt{3}+6}{9-12} = \frac{12+7\sqrt{3}}{-3}$ ✓

e) $\frac{x^2+x}{x^2-9} \times \frac{x^3-27}{x^2-x-2} = \frac{x(x+1)}{(x-3)(x+3)} \times \frac{(x-3)(x^2+3x+9)}{(x-2)(x+1)}$
 $= \frac{x(x^2+3x+9)}{(x+3)(x-2)}$ ✓

f) $f(x) = \begin{cases} -3 & \text{if } x \leq -2 \\ x^2 & -2 < x < 2 \\ 2x-3 & x \geq 2 \end{cases}$

i) $f(-1) + f(1) = (-1)^2 + 1^2 = 2$ ✓

ii) $f(a^2+2) = 2(a^2+2) - 3 = 2a^2 + 1$ ✓
always ≥ 2



Question 12

a) $G(x) = 4x - x^2$

i) $G(-3) = 4(-3) - (-3)^2 = -21$ ✓

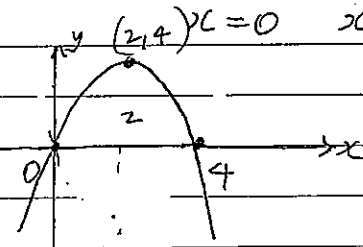
ii) $G(x) = 0 = 4x - x^2$

$0 = x(4 - x)$

$x = 0$ $x = 4$

✓ For both correct values

iii)



① - shape

① - intercepts and vertex

b) $2 \cos^2 \theta = 1$ $0 \leq \theta \leq 360^\circ$

$\cos^2 \theta = \frac{1}{2}$

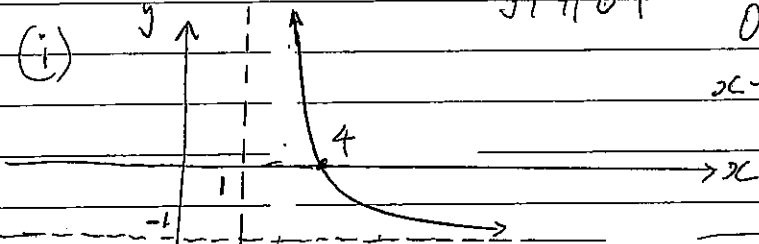
$\cos \theta = \pm \frac{1}{\sqrt{2}} \therefore \theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

c) $f(x) = \frac{3}{x-1} - 1$

x	0	4
y	-4	0

$0 = \frac{3}{x-1} - 1$

$x-1 = 3 \therefore x = 4$



shape - 1

asymptotes - 1

both x & y intercept

d) $\sin \theta = \frac{9}{5}$ and $\tan \theta > 0$

$\therefore \cos \theta = -\frac{\sqrt{24}}{5} = -\frac{2\sqrt{6}}{5}$

e) $(2\sqrt{3}+1)(3-\sqrt{3}) = a\sqrt{3} + b$

$6\sqrt{3} - \sqrt{3} + 3 - 6 = a\sqrt{3} + b$

$5\sqrt{3} - 3 = a\sqrt{3} + b$

$\therefore a = 5 \quad b = -3$

①

①

You may ask for extra writing paper if you need more space to answer question 12

Question 13

Name: _____

a)

$\tan 60^\circ = \frac{10}{x}$

$\sqrt{3} = \frac{10}{x} \therefore x = \frac{10}{\sqrt{3}}$ or $\frac{10\sqrt{3}}{3}$

b)

$AB^2 = 10^2 + 15^2 - 2 \times 10 \times 15 \times \cos 132^\circ$

$= 525.739...$

$AB = 22.929... = 22.93$ (ignore wrong rounding)

c) i) AC:

$A(3,0)$

$C(-2,8)$

$m_{AC} = \frac{8-0}{-2-3} = -\frac{8}{5}$

AC: $y - 0 = -\frac{8}{5}(x - 3)$

$5y = -8x + 24$

$\therefore 8x + 5y - 24 = 0$ shown

ii)

$B(x,y) \begin{cases} 3x + y - 9 = 0 \\ 4x - y + 16 = 0 \end{cases}$

either by substitution: $3(-1,12)$

$3x - 1 + 12 - 9 = 0$

$0 = 0$ True

B lies on $3x + y - 9 = 0$

$4(-1) - 12 + 16 = 0$ True

$\therefore B$ lies on $4x - y + 16 = 0$

OR

by solving

simultaneously

(1) $3x + y - 9 = 0$

(2) $4x - y + 16 = 0$

$7x + 7 = 0 \therefore x = -1$

$\therefore y = -3x - 1 + 9 = 12$ $B(-1,12)$

13c) iii) B(-1, 12)

$m_{AC} = -\frac{8}{5} \therefore m_{\perp line} = \frac{5}{8}$ ✓

$\therefore$ equation is $y - 12 = \frac{5}{8}(x - -1)$

$8y - 96 = 5x + 5$

in general form $\therefore 0 = 5x - 8y + 101$ ✓

a) $f(x) = y = \frac{1}{x^3}$

$f(-x) = \frac{1}{(-x)^3} = -\frac{1}{x^3} = -f(x) \therefore$ odd
①

e) $\frac{2}{3 - \sqrt{2}} + \frac{5}{3 + \sqrt{2}} = \frac{2(3 + \sqrt{2}) + 5(3 - \sqrt{2})}{(3 - \sqrt{2})(3 + \sqrt{2})} = \frac{21 - 3\sqrt{2}}{9 - 2} = \frac{21 - 3\sqrt{2}}{7}$ ✓

f) $\frac{a^{2x}}{a^{2y}} \times \frac{a^y}{a^2} = a^{2x+y-2y-2} = a^{2x-y-2} = a$ ✓

g) $\frac{y^{-1} + y}{1 + y^2} = \frac{\frac{1}{y} + y}{1 + y^2} = \frac{\frac{1 + y^2}{y}}{1 + y^2} = \frac{1}{y}$ ✓

Question 14

Name: _____

a) Prove $(\cos \alpha + \cot \alpha) \sec \alpha = 1 + \operatorname{cosec} \alpha$

LHS = $\left(\cos \alpha + \frac{\cos \alpha}{\sin \alpha} \right) \cdot \frac{1}{\cos \alpha}$

$= \cos \alpha \cdot \frac{1}{\cos \alpha} + \frac{\cos \alpha}{\sin \alpha} \cdot \frac{1}{\cos \alpha} = 1 + \frac{1}{\sin \alpha} = 1 + \operatorname{cosec} \alpha = \text{RHS}$
 $\therefore$ proven

b) $y = x^2$ $y = |4x - 3|$

(i) $x^2 = |4x - 3|$

case (1) $x^2 = 4x - 3$ OR case (2) $x^2 = -(4x - 3)$

$x^2 - 4x + 3 = 0$

$(x - 3)(x - 1) = 0$

$x = 3$

$y = 9$

$x = 1$

$y = 1$

$x^2 + 4x - 3 = 0$

$x = \frac{-4 \pm \sqrt{28}}{2} = \frac{-4 \pm 2\sqrt{7}}{2}$

$\therefore x = -2 \pm \sqrt{7}$

$x = -2 + \sqrt{7}$

$y = 11 - 4\sqrt{7}$

$x = -2 - \sqrt{7}$

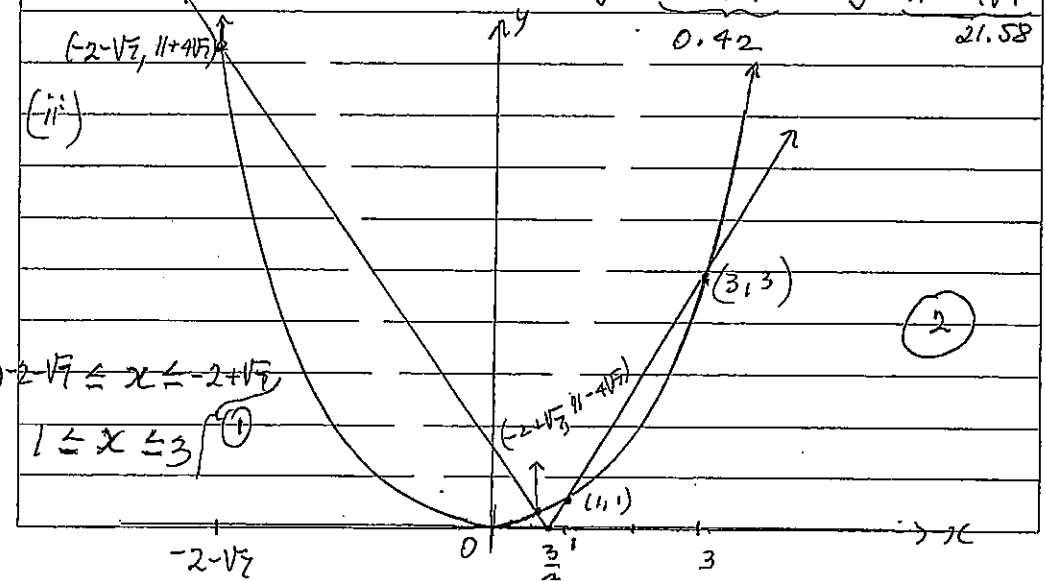
$y = 11 + 4\sqrt{7}$

$(-2 - \sqrt{7}, 11 + 4\sqrt{7})$

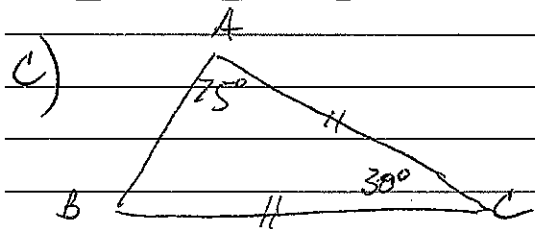
(ii)

ii) $-2 - \sqrt{7} \leq x \leq -2 + \sqrt{7}$

$1 \leq x \leq 3$ ①



Question 14



(i) let $AC = BC = a$

$$\therefore AB^2 = AC^2 + BC^2 - 2AC \cdot BC \cdot \cos 30^\circ$$

$$2^2 = a^2 + a^2 - 2a \cdot a \cdot \frac{\sqrt{3}}{2} \quad \checkmark$$

$$4 = a^2(2 - \sqrt{3})$$

$$a^2 = \frac{4}{2 - \sqrt{3}} \therefore a = \frac{2}{\sqrt{2 - \sqrt{3}}} = AC \quad \checkmark$$

only ∴ shown

ii) $\frac{\sin 75^\circ}{BC} = \frac{\sin 30^\circ}{2} \quad \checkmark$

$$\sin 75^\circ = \frac{BC \cdot \sin 30^\circ}{2} = \frac{\sqrt{2 - \sqrt{3}} \times \frac{1}{2}}{2}$$

$$= \frac{1}{\sqrt{2 - \sqrt{3}}} \cdot \frac{1}{2} = \frac{1}{2\sqrt{2 - \sqrt{3}}} \quad \checkmark$$

∴ shown

iii) $\text{Area} = \frac{1}{2} \times AB \times AC \times \sin 75^\circ$

$$= \frac{1}{2} \times 2 \times \frac{2}{\sqrt{2 - \sqrt{3}}} \cdot \frac{1}{2\sqrt{2 - \sqrt{3}}} \quad \checkmark$$

$$= \frac{1}{2 - \sqrt{3}}$$

OR $\text{Area} = \frac{1}{2} \times AC \times BC \times \sin 30^\circ$

$$= \frac{1}{2} \times \frac{2}{\sqrt{2 - \sqrt{3}}} \times \frac{2}{\sqrt{2 - \sqrt{3}}} \times \frac{1}{2} = \frac{1}{2 - \sqrt{3}} \quad \checkmark$$

iv) $\cos 75^\circ = \frac{AB^2 + AC^2 - BC^2}{2 \times AB \times AC} = \frac{2^2 + 0}{2 \times 2 \times AC} = \frac{4}{4 \times AC}$

$$= \frac{1}{AC} = \frac{\sqrt{2 - \sqrt{3}}}{2} \quad \checkmark$$

OR $\cos 75^\circ = \sqrt{1 - \sin^2 75^\circ} = \sqrt{1 - \left(\frac{1}{2\sqrt{2 - \sqrt{3}}}\right)^2} \quad \checkmark$

$$= \sqrt{1 - \frac{1}{4(2 - \sqrt{3})}} = \sqrt{1 - \frac{2 + \sqrt{3}}{4(4 - 3)}}$$

$$= \sqrt{\frac{4 - 2 - \sqrt{3}}{4}} = \frac{\sqrt{2 - \sqrt{3}}}{2} \quad \checkmark$$